Simulation Design for Week 3: *Right Message, Right Channel: A Social Marketing Simulation in Kaduna*
Course Code: MPH 8430_SDD_S4_01
Course: Social Marketing for Public Health Programmes
Topic: Marketing Channels in Public Health
Simulation Title: Right Message, Right Channel: A Social Marketing Simulation in Kaduna
Duration: Approx. 35-45 minutes (self-paced)
Format: Interactive, scenario-based simulation with decision points, hotspot activities, drag-and-drop classification, budget allocation sliders, branching decisions, and final stakeholder review.

SIMULATION SUMMARY

This simulation places learners in the role of a Health Communication Officer at the **State Primary Health Care Development Agency (SPHCDA)** in Kaduna, managing a 10-week immunisation campaign facing uneven uptake driven by misinformation, trust issues, and communication gaps. Across five decision points, learners identify barriers, select communication strategies, allocate budgets, adapt using monitoring data, and defend decisions before stakeholders. The simulation develops skills in evidence-based communication planning, social marketing, and adaptive public health decision-making in resource-constrained settings.

1. Simulation Purpose

This simulation develops learners' ability to design and adapt evidence-based communication strategies to improve immunisation uptake in a complex public health setting. It focuses on identifying barriers, selecting appropriate channels, allocating resources, and responding to monitoring data to support effective decision-making.

2. Learning Objectives
By the end of this simulation, learners will be able to:

- Identify and classify key communication barriers affecting immunisation uptake.
- Select appropriate communication strategies based on evidence and audience needs.
- Allocate communication budgets effectively under resource constraints.
- Adapt strategies using monitoring data to improve campaign outcomes.

3. SCENARIO SETUP

Screen Title:

Kaduna Immunisation Campaign Briefing

Visual Description:

- A modern **State Primary Health Care Development Agency (SPHCDA)** situation room in Kaduna State.
- Central digital dashboard showing immunisation indicators, LGA maps, coverage rates, and community feedback alerts.
- Large screen display of Kaduna State map highlighting Kaduna Metro, Zaria, and Ikara.
- Visual feeds of community outreach activities, radio broadcasts, mobile phone messaging, and town hall discussions.
- Campaign workspace with planning documents labelled *'10-Week Immunisation Campaign Strategy'*.
- Health officials analysing charts, discussing data, and reviewing field reports in real time.

Interactive Elements:

Learners can explore:

- Immunisation coverage trends across LGAs
- Community concerns and reported rumours
- Available communication channels
- Budget allocation overview

Situation:

You have just joined the State Primary Health Care Development Agency (SPHCDA) as the Health Communication Officer.

A 10-week immunisation campaign is underway, but uptake remains inconsistent across communities.

Your challenge is to identify communication barriers and develop a strategy that reaches the right audiences through the right channels.

Current Campaign Situation

Current Status

Issue	Status
Immunisation services	Available across LGAs
Uptake	Uneven across communities

Main challenges	Rumours, misinformation, and trust concerns
Available channels	Community mobilisation, radio, digital media, TV/outdoor advertising
Budget	₦180 million

Field Intelligence Dashboard

LGA	Coverage	Key Situation
Kaduna Metro	70%	Digital engagement strong, but misinformation is increasing online
Zaria	45%	Moderate uptake, trust-building required
Ikara	38%	High refusal linked to rumours and misinformation

Initial Mission Brief

Programme Director's Message

Before selecting communication channels, understand why communities are hesitant. The strategy must address the barriers affecting vaccine acceptance.

Transition to Decision Point 1 and screen 2

DECISION POINT 1: IDENTIFY COMMUNICATION BARRIERS

Interaction:

Hotspot + drag-and-drop activity

Visual Description:

- Kaduna SPHCDA communication analysis dashboard displayed on a large wall screen.

- Four interactive **field report cards** placed randomly across the workspace (not ordered):
  - Some pinned on a digital board
  - Some displayed on floating panels
- A separate **classification zone** with five labelled drop areas.
- Background visuals include:
  - Health workers conducting outreach in rural communities
  - Community meetings with traditional leaders
  - Radio studio recording health messages
  - Mobile phone notifications showing online vaccine discussions
- A side panel shows 'Campaign Performance Alerts' with uneven immunisation coverage across LGAs.

Learner Action:

1. Review each field report.
2. Identify the **primary communication barrier** in each case.
3. Drag each report into the correct category.

Field Reports (Randomised Presentation Order)

Field Report A:
'Parents are asking questions online, but programme responses are inconsistent.'

Field Report B:
'Some households are refusing MCV2 due to infertility rumours.'

Field Report C:
'Many rural households rely on radio and community discussions rather than online platforms.'

Field Report D:
'Community leaders want clarification about vaccine safety before encouraging families.'

Barrier Categories (Drop Zones):

- Misinformation / Trust Barrier
- Trusted Messenger Barrier
- Digital Communication Barrier
- Channel Access Barrier
- Not a Primary Communication Barrier

FEEDBACK SYSTEM (CRITICAL)

Correct Answers

Field Report A → Digital Communication Barrier

Feedback: Correct.
Online engagement requires timely and consistent responses. Weak coordination allows misinformation to spread.

Field Report B → Misinformation / Trust Barrier

Feedback: Correct.
The refusal is driven by false beliefs affecting confidence in vaccination.

Field Report C → Channel Access Barrier

Feedback: Correct.
Communication channels must match how communities actually receive information.

Field Report D → Trusted Messenger Barrier

Feedback: Correct.
Community leaders are key influencers in vaccine acceptance and must be actively engaged.

Incorrect Answers (System Feedback + Consequences)
Feedback (General Rule):

Incorrect classification. The selected barrier does not accurately reflect the underlying communication issue in the field report. Review how different types of communication barriers influence vaccine uptake.

Consequences System Triggers:

- Delayed unlock of Communication Strategy Phase
- Programme Director Warning Message activated

Learning Reset Prompt:

Learners are prompted to:

- Re-evaluate field reports
- Reconsider barrier types before progressing

Decision Point 1 Outcome

Strong Diagnosis:

- All or 3–4 correctly classified
  → Unlock: Communication Strategy Selection

Partial Diagnosis:

- 1–2 incorrect classifications
  → Proceed with warning + reduced effectiveness in next stage

Key Learning Message

Accurate identification of communication barriers is essential. Misdiagnosis leads to ineffective messaging strategies and reduced immunisation uptake.

Navigation:

Next → Screen 3: Communication Strategy Selection

DECISION POINT 2: SELECT COMMUNICATION STRATEGY

Interaction:

Branching choice-based decision

Visual Description:

- Kaduna SPHCDA communication command dashboard with strategy planning interface.
- A central decision panel displaying “Select Communication Strategy” with four selectable option cards.
- The background shows a split-view map of Kaduna State highlighting urban (Kaduna Metro, Zaria) and rural (Ikara and surrounding LGAs).
- Side panels display:
  - Rising misinformation alerts on mobile phones and social media feeds
  - Community engagement scenes with religious and traditional leaders
  - Radio studio broadcasting public health messages
  - Field health workers conducting household visits

- A system status bar shows campaign constraints: limited coordination capacity and uneven digital access across LGAs.

Situation Prompt:

You have reviewed field intelligence from Kaduna LGAs.

The Programme Director asks:

> 'We cannot use every communication channel equally. Based on the evidence, what strategy should guide the next phase of the campaign?'

Learner Action:

Select one communication strategy.

AVAILABLE STRATEGIC OPTIONS

Option A: Digital-First Strategy

- WhatsApp messaging
- SMS reminders
- Social media engagement
- Online misinformation response

Option B: Community Trust Strategy

- Community health workers
- Religious leaders
- Traditional leaders
- Household engagement

Option C: Visibility Strategy

- Television advertising
- Outdoor messaging
- Large public awareness campaigns

Option D: Integrated Adaptive Strategy

- Rural areas: CHWs, religious leaders, community discussions
- Urban areas: Digital platforms, SMS, social media engagement
- Statewide: Radio + coordinated messaging

FEEDBACK SECTION

Option A: Digital-First Strategy

Correct Answer:

❌ Not optimal as a standalone strategy

Feedback:

Strong for urban communication and rapid misinformation response, but it excludes communities with limited digital access and does not address deep trust issues.

Consequence:

- Kaduna Metro engagement improves
- Faster online response
- Rural hesitancy continues

Learning Message:

Digital tools improve speed but cannot replace offline trust-building mechanisms.

Option B: Community Trust Strategy

Correct Answer:

❌ Strong but incomplete alone

Feedback:

Highly effective for addressing vaccine hesitancy and trust barriers, especially in rural communities. However, it has limited reach and may not respond effectively to digital misinformation or urban communication needs.

Consequence:

- Improved acceptance in Ikara
- Stronger community trust
- Increased CHW workload

Learning Message:

Trust-based communication is powerful but requires system capacity and scalability.

Option C: Visibility Strategy

Correct Answer:

❌ Least effective standalone strategy

Feedback:

Increases awareness and visibility of the campaign but does not directly address misinformation, trust barriers, or behavioural change drivers.

Consequence:

- Increased public awareness
- Strong government visibility
- Persistent vaccine hesitancy

Learning Message:

Awareness alone does not lead to behaviour change.

Option D: Integrated Adaptive Strategy

Correct Answer:

✅ Most effective strategy

Feedback:

This approach aligns communication channels with audience needs across rural and urban settings. It combines trust-building, digital engagement, and mass communication for maximum effectiveness.

Consequence:

- Improved trust across LGAs
- Better coordination of messaging
- Increased monitoring complexity

Learning Message:

Effective public health communication requires adaptive, audience-specific strategies.

Navigation:

Next → Screen 4: Budget Allocation Decision

DECISION POINT 3: ALLOCATE COMMUNICATION BUDGET (₦180M)

Interaction:

Constrained Budget Allocation (Drag/Input-Based)

Visual Description:

- Kaduna SPHCDA financial and communication strategy dashboard.
- Central budget allocation interface showing ₦180 million total with interactive sliders for four communication channels.
- Side panel displays:
    - Measles outbreak alert in a neighbouring LGA (risk escalation warning)
    - Map of Kaduna State showing varying immunisation coverage intensity
    - Live campaign indicators (trust, reach, misinformation, visibility scores)
- Background visuals include:
    - Community outreach teams in rural settlements
    - Radio station broadcasting health messages
    - Mobile phones showing SMS and WhatsApp campaign updates
    - Billboards and TV campaign visuals across urban Kaduna

Situation Prompt:

A suspected measles cluster has been reported in a neighbouring LGA.

The Commissioner has requested a visible and effective response, but the campaign budget remains fixed at ₦180 million.

You must decide how to distribute resources across communication channels.

Budget Allocation Interface:

Channel	Allocation
Community mobilisation	₦____ million
Local radio	₦____ million
Digital media	₦____ million
TV and outdoor advertising	₦____ million
Total	**₦180 million**

BEFORE decision:

Channel Functions Guide

Community Mobilisation

- CHW outreach activities
- Religious and traditional leader engagement
- Household-level discussions
  Impact: Strong trust-building
  Risk: High workforce demand and fatigue

Local Radio

- Community radio programmes
- Phone-in health discussions
- Local language messaging
  Impact: Wide rural reach
  Risk: Limited personalised interaction

Digital Media

- WhatsApp messaging campaigns
- SMS reminders
- Social media monitoring and response
  Impact: Fast misinformation control
  Risk: Limited rural penetration

TV and Outdoor Advertising

- Billboards and posters
- Television health messaging
  Impact: High visibility and awareness
  Risk: Weak influence on entrenched hesitancy

AFTER decision:

System Outcome Model

BUDGET FEEDBACK (DYNAMIC OUTCOMES)

High Community Mobilisation Investment

Feedback:
Trust increases through direct engagement with communities.
Risk:
Potential CHW overload and operational strain.

High Digital Investment

Feedback:
Misinformation response improves rapidly.
Risk:
Rural populations may remain under-engaged.

High TV/Outdoor Investment

Feedback:
Campaign visibility increases significantly.
Risk:
Awareness improves, but behavioural change may remain limited.

High Local Radio Investment

Feedback:
Radio messaging significantly increases rural reach and improves access to health information in low-digital communities.

Risk:
Limited personalisation may reduce impact on deeply rooted misconceptions in some LGAs.

Balanced Investment

Feedback:
Resources are aligned with diverse communication needs across populations.

Learning Message:
Effective public health communication depends on strategic allocation, not maximum spending.

Transition:

All budget pathways feed into:

Week 6 Campaign Review Dashboard. Screen 5
The system evaluates whether adjustments are required based on performance outcomes.

Decision Point 4: Adapt Strategy Using Monitoring Data

Interaction:

Priority selection + Justification

Visual Description:

- Week 6 Kaduna SPHCDA monitoring dashboard displayed on a large wall screen in a strategy room.
- Central analytics interface showing real-time campaign indicators (coverage maps, trend graphs, alert flags).
- Split-screen comparison of urban vs rural performance across LGAs.
- Notification panels showing:
    - Reduced online misinformation alerts
    - Active offline community rumours
    - CHW workload warning indicators (fatigue alerts highlighted in red)
- Background visuals include:
    - CHWs conducting household visits in rural settlements
    - Radio stations broadcasting health messages
    - Mobile phones showing SMS and WhatsApp campaign responses
- A “Campaign Adjustment Panel” prompts the learner to modify strategy for Weeks 7–10.

Screen Title:

Week 6 Field Update — Campaign at a Crossroads

Situation Prompt

It is Week 6 of the campaign.

You are reviewing the latest monitoring dashboard when the programme director says:

> 'We are halfway through. The Commissioner wants a progress briefing. What are we changing for the second half of the campaign?'

Week 6 Monitoring Dashboard

Indicator	Status	Field Finding
Kaduna Metro coverage	Improving	Digital engagement is up, and urban parents are responding
Ikara & rural LGA uptake	Uneven	Offline concerns persist despite CHW visits
Online rumours	Reduced	Monitoring team responding faster
Offline rumours	Active	Community discussions still spreading hesitancy
CHW field reports	Concerning	Three teams reported fatigue and high workload
Budget remaining	40%	Must cover Weeks 7–10

Learner Task

Select the **priority adjustment** for the second half of the campaign.

Options

A. Increase Community Mobilisation

B. Expand digital misinformation response.

C. Increase radio and mass communication.

D. Rebalance communication channels.

FEEDBACK

Choice A: Increase Community Mobilisation

Feedback: Partially effective

Strengthening community engagement may improve trust in hesitant areas, but scaling CHW activity without additional support increases workload pressure.

Consequence:

- Rural community dialogue increases
- CHW fatigue worsens.
- Field teams request additional support.

Learning Message:

Trust-building must be matched with workforce capacity.

Choice B: Expand Digital Misinformation Response

Feedback:

Needs reconsideration

Digital response is already improving, but the main remaining challenge is offline hesitancy.

Consequence:

- Online confidence continues to improve
- Kaduna Metro coverage reaches ~80%
- Ikara uptake remains stagnant (38–40%)

Learning Message:

Strengthening already performing channels may not address the primary bottleneck.

Choice C: Increase Radio and Mass Communication

Feedback:

Partially effective

Radio improves reach and reduces pressure on field teams but cannot fully resolve trust-related barriers requiring interpersonal engagement.

Consequence:

- Message reach increases across LGAs
- Rural awareness improves
- Behaviour change remains slow
- CHW workload stabilises

Learning Message:

Mass communication supports awareness but is limited in driving deep behaviour change alone.

Choice D: Rebalance Communication Channels

Feedback:

Strong choice

TThis approach aligns with emerging field evidence by adjusting communication efforts across channels rather than over-relying on one.

Consequence:

- CHW workload becomes more sustainable
- Peer mobilisers activated in Ikara and 2 other LGAs
- Radio reinforces community messaging
- Coverage gap begins to improve

Learning Message:

Adaptive strategies outperform static approaches in dynamic public health environments.

System Update

Programme Director:

> 'This approach is harder to explain, but it reflects what the data is showing. '

Navigation:

Next → Screen 6: Final Campaign Evaluation

DECISION POINT 5: FINAL STAKEHOLDER REVIEW

Interaction:

BRANCHED RESPONSE + JUSTIFICATION

Visual Description:

- Formal government briefing room at the Kaduna State Ministry of Health.
- High-level stakeholder panel seated at a conference table (Commissioner, Programme Director, Finance representative, Governor's adviser).
- Large presentation screen showing campaign performance summary (coverage trends, trust indicators, communication reach).
- Slides displaying comparative analysis of communication strategies used across the campaign.
- Background elements include:
    - Official government branding and seals
    - Data dashboards projected behind the presenter
    - Printed briefing documents labeled "Final Campaign Review"
- Atmosphere reflects a high-stakes policy decision environment with executive-level scrutiny.

Screen Title:

Friday Briefing — The Commissioner's Office

Scenario

You are presenting campaign progress to:

- Commissioner of Health
- Programme Director
- Ministry of Finance representative
- Governor's Office adviser

Dynamic Opening Question:

The Commissioner responds based on prior decisions:

- **If CHWs were overused:**
  'Is this campaign sustainable with the current pressure on community teams?'
- **If digital was prioritised:**
  'Urban results are improving, but why are some rural communities still behind?'
- **If mass communication was prioritised:**
  'Messages have reached many people. Why are refusals still occurring?'

- **If integrated strategy was selected:**
  'Your approach is complex. Was using multiple channels worth the effort?'

Stakeholder Question:

The Commissioner asks:

> 'Why should we continue funding multiple communication channels when one channel may reach more people at lower cost?'

Learner Task:

Select ONE response and justify your communication strategy.

Learner chooses response:

A. Evidence-based integrated argument

B. Visibility-focused argument

C. Efficiency-focused argument

D. Trust-focused argument

RESPONSE OPTIONS

A. Evidence-Based Integrated Argument

Feedback:

✔ Strongest response

Different communities require different communication approaches. Digital channels enable rapid dissemination, while community-based channels address trust and misinformation. A multi-channel strategy ensures no key population group is excluded.

Outcome:

- Strong stakeholder confidence
- Approval to continue integrated funding approach

B. Visibility-Focused Argument

Feedback:

Limited effectiveness

Visibility increases awareness and demonstrates government action, but awareness alone does not address behavioural drivers of vaccine uptake.

Outcome:

- Moderate stakeholder support
- Request for stronger evidence of behavioural impact

C. Efficiency-Focused Argument

Feedback:

Moderate effectiveness

While cost efficiency is important, the lowest-cost approach may not achieve desired behaviour change across diverse populations.

Outcome:

- Stakeholders request clearer evidence of impact across LGAs

D. Trust-Focused Argument

Feedback:

Strong but incomplete

Trust-building is essential for vaccine acceptance, particularly in hesitant communities. However, trust strategies are most effective when combined with wider communication channels that ensure reach and reinforcement.

Outcome:

- Strong support for community engagement
- Concerns raised about scalability and system-wide coverage

ENDING 1: Integrated Channel Strategy — Campaign Succeeds

Trigger Conditions:

Learner:

- Selects an integrated communication strategy (DP2)
- Applies balanced budget allocation (DP3)
- Uses monitoring data to adapt strategy effectively (DP4)
- Provides evidence-based justification (DP5)

Campaign Outcome:

- Kaduna Metro coverage reaches **82%**
- Ikara uptake improves from **38% → 61%**
- Online rumours significantly reduced
- Offline hesitancy addressed through peer mobilisers and CHWs
- CHW workload stabilised through rebalancing of tasks

Learning Message:

Successful public health communication depends on aligning channels with audience needs and continuously adapting based on evidence. Effectiveness comes from *strategic integration*, not channel quantity.

ENDING 2: High Visibility, Limited Behaviour Change

Trigger Conditions:

Learner:

- Prioritises TV/outdoor communication heavily
- Focuses mainly on awareness creation

- Underinvests in trust-building strategies (community engagement / CHWs)

Campaign Outcome:

- Awareness increases across all LGAs
- Kaduna Metro coverage improves moderately
- Ikara increases from **38% → 44%**
- Strong campaign visibility achieved
- Persistent hesitancy remains in rural communities

Learning Message:

High visibility improves awareness, but awareness alone does not lead to behaviour change. Without trust and engagement, hesitancy remains.

ENDING 3: Strong Trust, Limited Scale

Trigger Conditions:

Learner:

- Heavily prioritises community mobilisation
- Achieves strong trust-building outcomes
- Does not adequately scale digital or mass communication channels

Campaign Outcome:

- Ikara shows strong improvement in acceptance
- Vaccine refusal rates decline in trust-sensitive communities
- Community confidence increases significantly
- Overall coverage grows slowly due to limited reach
- Urban expansion remains moderate

Learning Message:

Trust is essential for behaviour change, but sustainable public health impact requires scalable communication systems that extend beyond interpersonal engagement.